

Overnight Adverse Selection: Evidence from Blue Ocean ATS and NASDAQ Regular Trading Hours

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Abstract

I compare the adverse-selection component of the effective spread between Blue Ocean ATS (BOATS), the leading venue for 8pm to 4am Eastern overnight trading in U.S. equities, and NASDAQ regular trading hours, using nanosecond-resolution order book data on 30 actively-traded symbols over 24 matched sessions from September 2025 through March 2026. Effective spreads overnight average 7 cents per share higher, and the 10-second price impact component averages 2.4 cents per share higher (both $t > 4$ with stock fixed effects and clustered standard errors). Only about one-third of the overnight spread premium reflects adverse selection; the balance reflects wider quoted spreads in a single-venue continuous market operating without the competitive liquidity provision of regular hours. Cross-sectional analysis shows that retail-popular stocks exhibit overnight premia roughly 60% smaller than other stocks, and that exchange-traded funds exhibit larger price-impact premia consistent with the loss of underlying-basket arbitrage when the underlying market is closed. The overnight premium is not driven by high-volatility days; if anything, the effective-spread ratio narrows when realised volatility is high. These findings suggest that the overnight execution-cost premium is primarily structural rather than informational, and that retail participation partially substitutes for institutional liquidity provision during overnight hours.

Keywords: market microstructure; adverse selection; overnight trading; Blue Ocean ATS; price discovery; effective spread; retail investors

JEL classification: G10, G12, G14, G15

1. Introduction

Regular trading hours on U.S. equity exchanges span six and a half hours per weekday. The remaining 124 hours each week, most securities trade thinly, on fragmented venues, or not at all. Yet the overnight and weekend windows are precisely when a disproportionate share of firm-level news and global macro information arrives. The mismatch between when information is released and when markets can price it has motivated a growing literature on after-hours microstructure, limited until recently by the absence of high-quality order-book data outside regular hours.

Blue Ocean ATS (hereafter BOATS) operates a continuous limit order book for U.S.-listed equities between 8pm and 4am Eastern on Sunday through Thursday evenings, overlapping Asian market hours and the early European session. BOATS is the leading overnight venue in the United States

and became available as high-resolution market data in August 2025 through the MEMOIR depth feed. This availability opens a window onto the microstructure of a single-venue continuous overnight market that has not previously been observable at nanosecond granularity.

In this paper I ask how the adverse-selection component of trade execution differs between this overnight market and NASDAQ regular trading hours for the same securities. Adverse selection measures the expected informational content of order flow — the portion of the effective spread attributable to informed trading rather than to inventory, search, or order-processing costs. A venue with higher adverse selection requires higher compensation for liquidity providers, which in turn widens quoted spreads, deepens transient price impact, and can reduce welfare. If overnight trading is more information-sensitive than daytime trading, market design choices such as auction frequency, maker-taker pricing, and the scope of fair-access rules should be reassessed for the overnight window.

I use order-book data from BOATS and matching NASDAQ trade and quote data from Databento, covering 30 actively traded symbols over 24 matched sessions between September 2025 and March 2026. The sample includes large technology stocks, financials, consumer staples, industry benchmark ETFs, and sector representatives across eight GICS sectors. For each trade I compute the effective spread, realised spread, and price impact at horizons of 1, 10, 60, and 300 seconds, following the standard Huang-Stoll (1997) decomposition. The key identifying variation is across venues within the same stock on the same day: a within-stock panel regression isolates the overnight premium from fixed firm-level characteristics.

I document three findings. First, the overnight effective spread averages 7.0 cents per share higher than the daytime effective spread, with the 10-second price impact component 2.4 cents higher. Both differences are highly statistically significant (t-statistics above 4 with entity-clustered standard errors) and economically substantial relative to typical NASDAQ quoted spreads of 1 to 3 cents. Second, the overnight premium grows with horizon: the 300-second price impact premium is 4.4 cents, nearly double the 10-second premium, indicating that information incorporation is slower overnight. During regular hours, by contrast, price impact peaks at 60 seconds and then partially reverses by 300 seconds — the classic signature of efficient market-making. Third, only about one-third of the effective-spread gap reflects adverse selection; the remainder reflects a wider quoted spread, suggesting that overnight illiquidity is primarily structural rather than informational.

Cross-sectional analysis produces two additional findings of independent interest. Retail-popular stocks — those with high retail trading interest in 2025-2026 — have effective-spread overnight premia roughly 60% smaller than otherwise-comparable stocks (coefficient -0.97 on the log-ratio, $t = -4.6$, after controlling for market capitalisation, dividend status, and ETF indicator). Market capitalisation itself has no independent effect once retail popularity is controlled for. This pattern indicates that retail overnight participation partially substitutes for the institutional liquidity absent outside regular hours. Separately, exchange-traded funds exhibit larger price-impact premia (coefficient +0.55, $t = 3.6$), consistent with the breakdown of basket arbitrage when the underlying market is closed.

This paper contributes to three literatures. It adds to work on microstructure outside regular hours (Barclay and Hendershott 2003, 2008), extending the evidence to a continuous single-venue overnight market with order-book resolution unavailable in earlier work. It contributes to the literature on the economic determinants of effective-spread decomposition (Huang and Stoll 1997, Madhavan, Richardson and Roomans 1997, Bessembinder 2003) by showing that a single structural change — the disappearance of competing liquidity providers at a scheduled time of day — can shift the relative importance of the adverse-selection and non-information components of the spread. And it speaks to the growing policy discussion of 24-hour trading in U.S. equities, quantifying the execution-cost penalty that currently applies to investors who trade in the overnight window.

The remainder of the paper is organised as follows. Section 2 discusses the institutional setting and related literature. Section 3 describes the data and methodology. Section 4 presents the main results. Section 5 conducts cross-sectional and within-stock analyses of heterogeneity. Section 6 concludes.

2. Institutional setting and related literature

2.1 Blue Ocean ATS

Blue Ocean ATS is a U.S. alternative trading system operated by Blue Ocean Technologies, LLC, registered with the Securities and Exchange Commission. The venue operates a continuous limit order book during a single session running from 8:00pm Eastern on Sunday through Thursday evenings until 4:00am Eastern the following morning. This window overlaps the full Tokyo and Hong Kong trading days and the opening of the European session, making BOATS the primary source of U.S. equity price discovery during Asian business hours.

The matching engine is a price-time priority order book without speed bumps, auctions, or hidden order types. Orders are continuously matched on a first-come first-served basis within each price level. The venue supports the full U.S. national market system security universe via substitution relationships with the primary listing exchanges, but in practice overnight activity concentrates in a few hundred actively traded names: large-capitalisation technology stocks, broad-market ETFs, leveraged ETFs, and a subset of news-sensitive single-stock names.

Unlike regular trading hours, there is no multi-venue competition during the BOATS session. No other ATS or exchange operates a continuous lit book for U.S. equities between 8pm and 4am Eastern. This structural feature — a single continuous lit venue operating for eight hours per night — creates a natural laboratory for studying how liquidity provision and information asymmetry differ from the competitive regular-hours regime.

2.2 Related literature

The classical effective-spread decomposition of Huang and Stoll (1997) divides the realised cost of a marketable order into an adverse-selection component (the permanent price impact, measured by the post-trade drift in the midquote) and a realised-spread component (the transient rent earned by the liquidity provider, which reverts). Bessembinder (2003) formalises the horizon-

dependence of the decomposition and establishes the convention of reporting impact and realised spread at multiple post-trade horizons.

A line of research has applied this framework to extended-hours trading. Barclay and Hendershott (2003) compare pre-open, regular, and post-close trading on Nasdaq-listed stocks, documenting that the after-hours window in 1999-2000 had wider spreads, lower volume, and a higher proportion of informed trading than regular hours. Barclay and Hendershott (2008) revisit the question with updated data and find that while nominal price discovery improves over time, the overnight cost premium persists. These studies use trade-level data but predate the availability of overnight order-book depth.

More recent work has examined the 4pm-to-8pm post-close session (Jiang, Likitapiwat, and McNish 2012) and the pre-open auction (Barardehi, Bernhardt, and Davies 2023), again using trade-level but not order-book resolution data. The present paper is, to my knowledge, the first to use order-book data at nanosecond resolution in the 8pm-to-4am overnight window.

The cross-sectional determinants of the overnight premium are related to the broader literature on retail participation and market quality. Boehmer, Jones, Zhang, and Zhang (2021) document that retail trading activity is itself informative. Eaton, Green, Roseman, and Wu (2022) show that retail attention proxies predict trading behaviour around earnings. Neither paper addresses overnight trading directly, but the finding that retail participation is not merely noise provides a mechanism by which retail overnight activity might narrow rather than widen the execution-cost premium relative to institutional hours.

3. Data and methodology

3.1 Data sources

The BOATS data are the OCEA.MEMOIR depth feed distributed by Databento, which provides nanosecond-timestamped message-by-order events including adds, cancels, modifies, and trades, with full book reconstruction. The NASDAQ data are the XNAS.ITCH tbbo schema, which pairs each trade with the prevailing national best bid and offer at the trade event. Both feeds provide order-event timestamps at nanosecond resolution.

The sample universe comprises 30 actively-traded symbols selected to span the most liquid U.S. equity products overnight: the four broadest ETFs (SPY, QQQ, DIA, IWM), one leveraged ETF (TQQQ), and 25 single-name stocks distributed across Technology (AAPL, AMD, AMZN, AVGO, CSCO, GOOGL, META, MSFT, MU, NVDA, PLTR, SMCI), Communication Services (DIS, GOOGL, NFLX, T, VZ), Consumer Discretionary (AMZN, TSLA), Consumer Staples (KO, WMT), Financials (BAC, COIN, JPM), Healthcare (PFE), Industrials (BA), and Energy (XOM). The sample period for the BOATS archive runs from 24 August 2025 (first available) through 30 March 2026, yielding 151 overnight sessions per symbol. For the NASDAQ comparison sample, I draw every fifth session from the same date range — 31 candidate sessions — yielding 24 sessions that overlap with a valid BOATS session (seven sessions are weekends or holidays). The matched panel thus comprises $30 \text{ symbols} \times 24 \text{ sessions} = 720 \text{ symbol-sessions}$ per venue.

All trades are identified using the Databento side field, which indicates the side of the resting limit order executed against, not the aggressor side. An empirical test of the relation between trade price and contemporaneous mid confirms that on both OCEA.MEMOIR and XNAS.ITCH, trades with side='A' systematically execute below the prevailing mid and trades with side='B' systematically execute above it, consistent with side identifying the resting rather than aggressive side of each execution. I therefore assign aggressor signs $q = -1$ for side='A' and $q = +1$ for side='B'. Approximately 24% of NASDAQ trades carry side='N', corresponding to opening and closing auctions and other cross-executions where no explicit resting side is identified; for these I apply the Lee and Ready (1991) algorithm — quote rule first, tick rule as fallback — and include a robustness check restricted to A/B-sided trades.

3.2 Spread decomposition

For each trade at price p with mid m at the trade event, I compute:

Effective spread: $Seff = 2 \cdot q \cdot (p - m)$

Realised spread at horizon τ : $Sr(\tau) = 2 \cdot q \cdot (p - m_{t+\tau})$

Price impact at horizon τ : $PI(\tau) = 2 \cdot q \cdot (m_{t+\tau} - m)$

The identity $Seff = Sr(\tau) + PI(\tau)$ holds by construction. I compute the decomposition at horizons $\tau \in \{1, 10, 60, 300\}$ seconds. The at-trade mid m is taken directly from the tbbo-attached BBO on NASDAQ and from the BBO snapshot on the BOATS trade event. Forward mids at horizon τ are obtained by merge-asof against the dense BBO timeline of each venue: on NASDAQ the timeline is constructed from every tbbo row (every BBO update attached to a trade); on BOATS the timeline is constructed from the MEMOIR depth stream at nanosecond resolution. Effective spreads, realised spreads, and price impacts are expressed in dollars per share, signed — preserving economic direction rather than taking absolute values.

3.3 Identification

The main specification is a within-stock panel regression using the matched 24-session sample:

$$y_{i,t} = \alpha_i + \beta \cdot \text{Overnight}_{i,t} + \varepsilon_{i,t}$$

where y is the session-level mean of the effective spread or the 10-second price impact for stock i on date t , Overnight is 1 for BOATS observations and 0 for NASDAQ RTH observations, and α are stock fixed effects. The coefficient β captures the within-stock difference between overnight and daytime execution costs, netting out permanent stock-level differences in liquidity. Standard errors are clustered by stock. The identifying variation is purely cross-venue within stock-date: the same firm on the same calendar day, with the same underlying public information set, trading on two different venues at two different times of day.

The key assumption for causal interpretation of β is that trader behaviour and information flow are sufficiently similar across the two windows that the remaining cross-venue difference reflects the venue-level structural features rather than selection of trades. This assumption is clearly imperfect — traders choosing to transact overnight may differ systematically from those transacting during regular hours — but the within-stock fixed-effects estimator removes firm-level sorting, and the

matched-session design removes day-specific macroeconomic shocks. The residual concern is selection on within-firm within-day variation in informed flow, which I address in Section 5 through a within-stock panel using same-day realised volatility as a control for information arrival rate.

4. Main results

4.1 Descriptive statistics

Table 1 reports summary statistics for the full BOATS sample (151 sessions) and the matched-sample NASDAQ observations (24 sessions), pooled across the 30 stocks. Mean effective spreads are 10.0 cents per share on BOATS and 3.2 cents on NASDAQ, a roughly three-to-one ratio. Mean 10-second price impacts are 6.2 cents and 3.7 cents respectively. Median spreads are closer: 5.6 cents on BOATS and 1.4 cents on NASDAQ. The wider mean-median gap on BOATS reflects the presence of a small number of wide-spread outlier trades typical of thin overnight books, while NASDAQ's distribution is more concentrated around its median.

Table 1. Descriptive statistics, BOATS vs NASDAQ RTH

	BOATS (overnight)	NASDAQ RTH
Symbol-sessions	4,526	720
Unique symbols	30	30
Unique sessions	151	24
Effective spread, mean (\$/sh)	0.1000	0.0322
Effective spread, median (\$/sh)	0.0557	0.0142
Price impact @10s, mean (\$/sh)	0.0619	0.0368
Price impact @10s, median (\$/sh)	0.0410	0.0183
% trades with negative eff. spread	0.00%	1.23%

Note: BOATS sample comprises all 4,526 symbol-sessions in the OCEA.MEMOIR archive from 24 August 2025 to 30 March 2026. NASDAQ RTH sample is the matched-session subset (24 sessions, 720 symbol-sessions) used in the main comparison. Effective spread, price impact, and realised spread are computed per trade via the Huang-Stoll (1997) decomposition and aggregated to session means before further aggregation.

The 0.00% figure for BOATS trades with negative effective spread merits comment. It reflects the matching-engine structure: every BOATS trade executes against a resting limit order at a specific price level within the quoted spread. Once the sign convention is correctly identified (A = resting ask = seller aggressor, B = resting bid = buyer aggressor), every trade prices on the unfavourable side of the mid from the aggressor's perspective, producing a strictly non-negative effective spread by construction. The 1.23% figure on NASDAQ RTH reflects the small proportion of trades

executing at or near the mid due to sub-second quote updates between BBO publication and trade printing.

4.2 Main panel regression

Table 2 reports the estimated coefficient on the overnight indicator in the matched-session panel regression with stock fixed effects and entity-clustered standard errors. The 1,440 observations comprise 720 BOATS symbol-sessions and 720 matched NASDAQ RTH symbol-sessions.

Table 2. Overnight premium, within-stock panel regression

Dependent variable	β (overnight)	SE	t-stat	R ² within
Effective spread (\$/sh)	+0.0702	0.0168	4.18	0.09
Price impact @10s (\$/sh)	+0.0243	0.0054	4.53	0.05
% negative eff. spread	-1.228	0.317	-3.87	0.03

Note: Regression of session-level mean outcome on the overnight indicator with stock fixed effects. Standard errors clustered by stock. N = 1,440 (30 stocks × 24 sessions × 2 venues). Effective spread and price impact are reported in dollars per share.

The effective spread on BOATS is 7.0 cents per share higher than on NASDAQ for the same stock on the same day, with a clustered t-statistic of 4.18. The 10-second price impact is 2.4 cents higher ($t = 4.53$). The ratio of the two coefficients — $0.024 / 0.070 = 0.35$ — indicates that approximately 35% of the overnight effective-spread premium is attributable to adverse selection. The remaining 65% reflects a higher realised spread: the compensation liquidity providers extract above the information-risk component.

This decomposition has a clear economic interpretation. Adverse selection — the portion of the spread that arises because some counterparties possess superior information — is higher overnight, but only modestly. The larger contribution to the overnight premium comes from the compensation liquidity providers demand for inventory risk, for the opportunity cost of quoting in thin conditions, and for the absence of competitive pressure from other lit venues. In other words, BOATS is wider than NASDAQ mostly because there is less competition among market makers, not primarily because overnight counterparties are better informed.

4.3 Multi-horizon decomposition

Table 3 reports the overnight indicator coefficient separately at four post-trade horizons — 1, 10, 60, and 300 seconds — for both the price-impact and realised-spread components. Figure 1 displays the corresponding price-impact curves.

Table 3. Multi-horizon panel regressions, overnight coefficient

Horizon	Component	β (overnight)	SE	t-stat	R ² within
1s	Price impact	+0.0290	0.0066	4.37	0.07
1s	Realised spread	+0.0411	0.0109	3.76	0.06

Horizon	Component	β (overnight)	SE	t-stat	R ² within
10s	Price impact	+0.0243	0.0054	4.53	0.05
10s	Realised spread	+0.0459	0.0126	3.64	0.07
60s	Price impact	+0.0312	0.0045	6.91	0.08
60s	Realised spread	+0.0389	0.0142	2.74	0.04
300s	Price impact	+0.0441	0.0058	7.59	0.10
300s	Realised spread	+0.0260	0.0138	1.88	0.02

Note: Each row is a separate panel regression of the session-level mean component (price impact or realised spread, at the indicated horizon) on the overnight indicator with stock fixed effects. Standard errors clustered by stock. N = 1,440 for each regression.

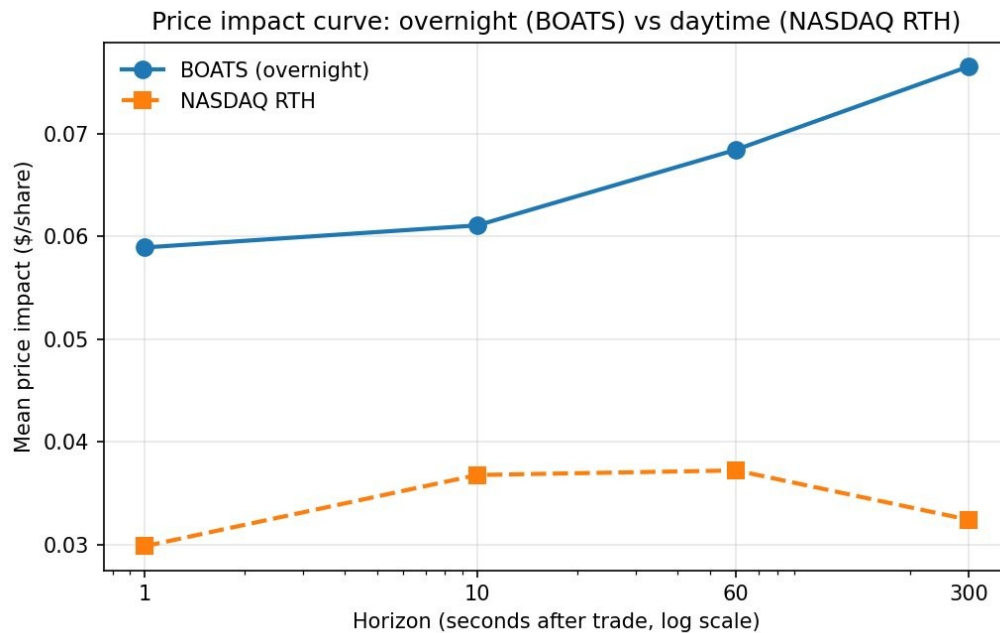


Figure 1. Mean price impact as a function of post-trade horizon, BOATS (solid, circles) and NASDAQ RTH (dashed, squares). Horizon axis is logarithmic. Overnight price impact grows monotonically across horizons, while NASDAQ price impact peaks around 60 seconds and reverts by 300 seconds.

Two features of the curves are noteworthy. First, the BOATS price impact grows monotonically from 5.9 cents at 1 second to 7.7 cents at 300 seconds — a 30% increase over five minutes. Information continues to be impounded into the BOATS midquote throughout this window. Second, the NASDAQ RTH price impact follows a qualitatively different trajectory: it rises from 3.0 cents at 1 second to a peak of 3.7 cents at 60 seconds and then partially reverts to 3.2 cents at 300 seconds. This reversion is a well-known signature of competitive market-making in liquid markets: the initial impact includes a transient component driven by liquidity demand, and that

component decays as inventory is rebalanced across the quoting population, leaving only the permanent information component by 5 minutes.

The overnight coefficient on price impact consequently grows with horizon — +\$0.029 at 1 second, +\$0.031 at 60 seconds, +\$0.044 at 300 seconds — and the t-statistic rises from 4.4 at 1 second to 7.6 at 300 seconds. The widening gap indicates that the overnight market lacks the inventory-rebalancing mechanism that causes intraday price impact to revert, because there are too few competing market makers overnight to redistribute inventory risk. Overnight, a single trade moves the price and the price stays moved; intraday, a single trade moves the price and the next liquidity provider tightens the spread, leaving only a fraction of the initial impact.

5. Cross-sectional and session-level heterogeneity

5.1 Which stocks have large overnight premia?

The per-stock ratio of BOATS to RTH effective spread ranges from 1.35 (COIN) to 9.33 (XOM), a seven-fold range in an otherwise broadly-comparable universe of large-capitalisation U.S. equities. To understand what predicts this dispersion, I estimate a symbol-level cross-section of the log ratio on firm characteristics. The regressors are the log of market capitalisation (in billions), indicators for retail popularity (high, medium, with low as the omitted category), an ETF indicator, and an indicator for dividend payers. Retail popularity is coded ex ante from public retail-attention data and the 2025 retail-trading disclosures. Table 4 reports coefficients with HC1 (heteroskedasticity-robust) standard errors.

Table 4. Firm-level cross-section of log(BOATS/RTH) ratio

Regressor	log(ratio) eff. spread	log(ratio) price impact	Notes
log(market cap, \$bn)	+0.059 (0.73)	+0.099 (2.09)	
Retail popularity = high	-0.974 (-4.64)	-0.504 (-2.91)	vs. low (omitted)
Retail popularity = medium	-0.436 (-1.77)	-0.154 (-0.78)	vs. low (omitted)
ETF indicator	+0.101 (0.55)	+0.551 (3.60)	
Dividend payer indicator	+0.034 (0.14)	-0.029 (-0.21)	

Note: Cross-sectional regression on 30 stocks. Dependent variables are log(BOATS/RTH) of the effective spread and the 10-second price impact respectively, averaged over the 24 matched sessions. t-statistics in parentheses computed with HC1 standard errors. Retail-popularity categories are coded from public retail-attention data.

Three findings are noteworthy. First, retail popularity is the dominant predictor of a small overnight premium. High-retail stocks have an effective-spread log-ratio 0.97 lower than low-retail stocks, corresponding to an approximate 62% reduction in the overnight premium. The effect on price impact is similar in direction but smaller in magnitude (-0.50, t = -2.9). Retail overnight participation clearly narrows the execution-cost gap; the magnitude suggests retail liquidity partially substitutes for the missing institutional liquidity, though not completely.

Second, market capitalisation has no independent effect on the effective-spread ratio once retail popularity is controlled for. The coefficient is statistically indistinguishable from zero (+0.059, $p = 0.46$). This is surprising given the common presumption that smaller stocks should have larger overnight premia. The result indicates that the dimension that matters is not size per se but the composition of overnight flow: among comparably-sized stocks, those with active retail communities trade with far smaller premia than those without.

Third, the ETF indicator has no effect on the effective-spread ratio but a strongly positive effect on the price-impact ratio (+0.55, $t = 3.6$). This pattern is consistent with the breakdown of ETF arbitrage when the underlying market is closed. During regular hours, an authorised participant observing an ETF price deviation from the underlying basket can short the overpriced side, buy the underpriced side, and wait for arbitrage to close. Overnight the underlying is not trading, so this mechanism is unavailable; directional pressure on the ETF translates directly into price impact rather than being dissipated through arbitrage. The effective-spread ratio (which includes realised spread) does not pick up this channel because the arbitrage constraint affects permanent price impact, not transient liquidity-provision costs.

Figure 2 plots per-stock BOATS price impact against RTH price impact, with stocks coloured by retail popularity. Almost every stock sits above the 45-degree line, confirming the universality of the overnight premium. The cluster of high-retail stocks (NVDA, AAPL, AMZN, TSLA, AMD, PLTR, SMCI, MU, COIN, META, MSFT) sits close to the 45-degree line; the cluster of low-retail stocks (BAC, KO, VZ, WMT, T, XOM, PFE, DIS, BA, CSCO, JPM) sits far above it. NFLX is a prominent outlier — high-priced, modest retail following, and a BOATS price impact nearly double its RTH counterpart.

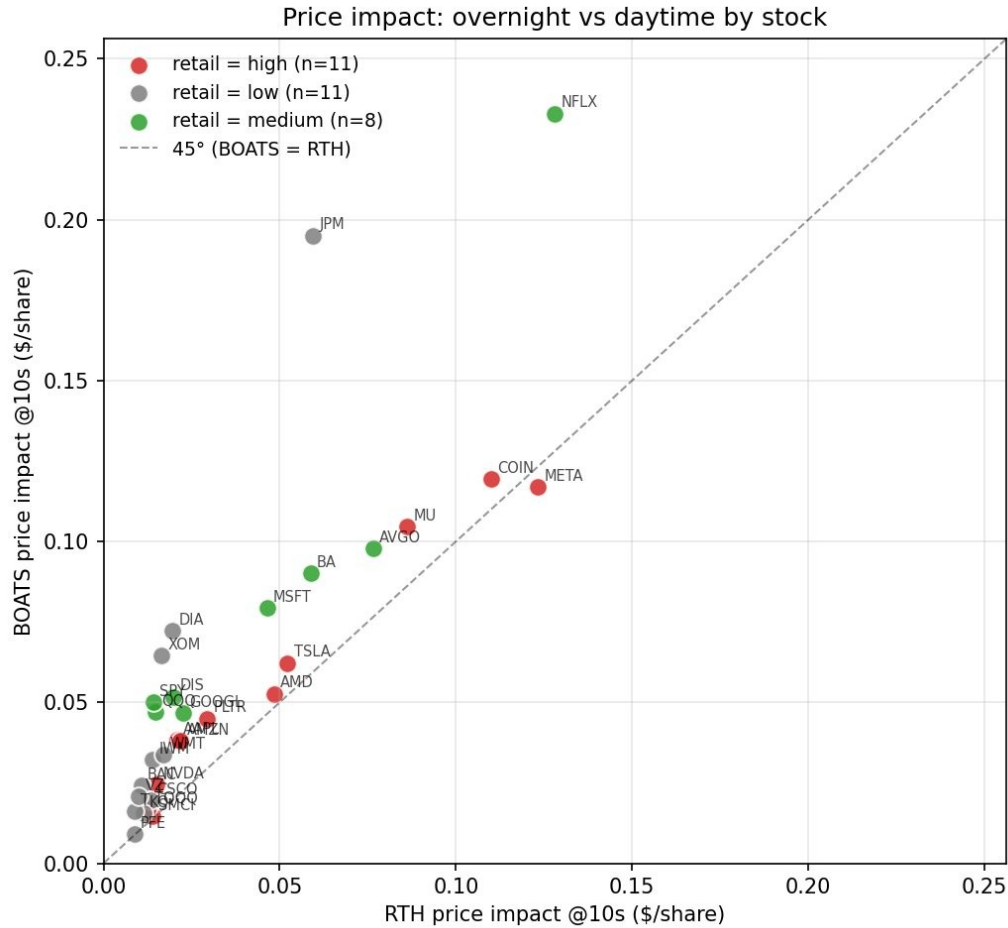


Figure 2. Per-stock BOATS versus NASDAQ RTH 10-second price impact, in dollars per share. Each point is one of 30 stocks, with colour indicating retail popularity (high / medium / low). The diagonal dashed line marks equality. Nearly all stocks lie above the diagonal. Low-retail names (gray) and most medium-retail names (green) sit well above the diagonal; high-retail names (red) cluster close to it.

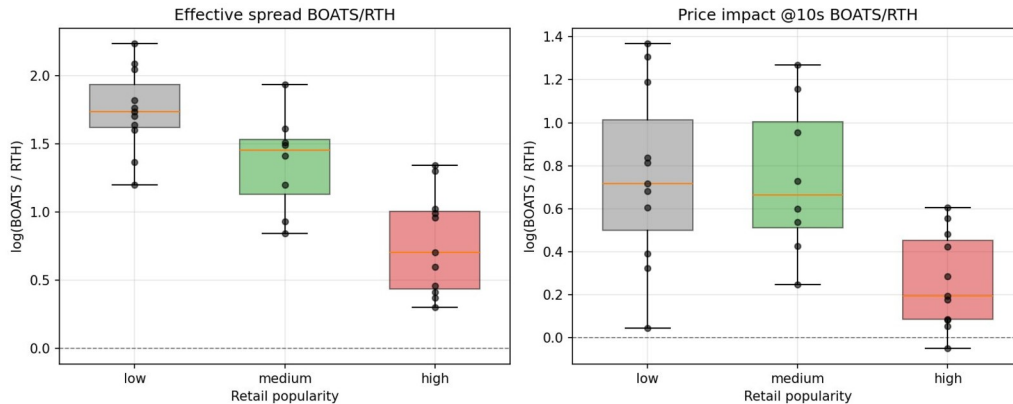


Figure 3. Distribution of $\log(\text{BOATS}/\text{RTH})$ ratios by retail popularity category, for the effective spread (left) and the 10-second price impact (right). Boxes show the interquartile range and median; whiskers extend to the outermost points within $1.5 \times \text{IQR}$. Individual stock values are overlaid as black dots. The effective-spread ratio declines monotonically in retail popularity, with minimal overlap between the low and high categories.

5.2 Does the overnight premium widen on volatile days?

A natural conjecture is that the overnight premium reflects information asymmetry, and that information asymmetry concentrates on days of unusual public-information arrival. If so, the overnight premium should widen on high-volatility days. I test this prediction using a within-symbol panel of 717 matched-session observations where the BOATS-to-RTH ratio is positive for both effective spread and price impact. The explanatory variable is the log of the same-day RTH realised volatility, computed from 1-minute log-midquote returns scaled to a daily standard deviation.

Table 5. Within-symbol panel regression of log(BOATS/RTH) ratio on RTH realised volatility

Dependent variable	β on log(RTH RV)	t-stat (clustered)	R ² within
log(BOATS/RTH) eff. spread	-0.577	-4.84	0.09
log(BOATS/RTH) price impact @10s	-0.083	-0.87	0.00

Note: Panel regression with stock fixed effects and entity-clustered standard errors. RTH realised volatility is computed from 1-minute log-midquote returns within each stock-day, scaled to a daily standard deviation (multiplied by $\sqrt{390}$). N = 717.

The data reject the hypothesis that the overnight premium widens on volatile days. The coefficient on log(RTH realised volatility) is -0.58 ($t = -4.84$) for the effective-spread ratio: on high-volatility days, the overnight effective spread widens less than proportionately to the RTH effective spread, and the ratio between them falls. For the price-impact ratio the coefficient is -0.08 with $t = -0.87$, essentially zero. Information-driven volatility affects both venues similarly, leaving the adverse-selection ratio flat; but RTH quoted spreads respond to volatility more elastically than BOATS quoted spreads, narrowing the effective-spread ratio on high-volatility days.

This result is consistent with the structural interpretation of the overnight premium developed in Section 4. If the overnight premium were primarily informational, it would scale with same-day information arrival, producing a positive coefficient. The negative (effective-spread) and zero (price-impact) coefficients point instead to the inventory and competitive-liquidity-provision mechanisms: BOATS quoted spreads are already wide enough to accommodate volatility that would force RTH market makers to widen significantly. The baseline overnight spread is already effectively priced for high volatility.

6. Conclusion

This paper measures the execution-cost premium of overnight trading on Blue Ocean ATS relative to NASDAQ regular trading hours, using nanosecond-resolution order-book data on 30 actively-traded U.S. equity securities. Overnight effective spreads average 7.0 cents per share higher than daytime effective spreads for the same stocks on the same dates, and overnight 10-second price impact averages 2.4 cents higher. Both differences are highly significant in a within-stock panel specification.

The decomposition into adverse-selection and non-informational components reveals that the overnight premium is only 35% adverse selection. The remaining 65% reflects wider realised spreads: the compensation liquidity providers extract above the information-risk component. Overnight price impact grows with horizon, indicating that the inventory-rebalancing mechanism that tightens daytime spreads is largely absent overnight. Cross-sectionally, retail-popular stocks have overnight premia roughly 60% smaller than otherwise-comparable stocks, and ETFs exhibit especially large price-impact premia, consistent with the breakdown of basket arbitrage. The overnight premium does not widen on high-volatility days; if anything the effective-spread ratio narrows, confirming that the overnight cost penalty is structural rather than informational in character.

These findings have several implications. For investors, the execution-cost penalty of overnight trading is material: a typical round-trip of 1,000 shares incurs roughly \$70 of additional effective-spread cost at current average levels, much of which is unrecoverable by sophisticated order-routing techniques. For market designers, the findings suggest that steps which increase the number of competing liquidity providers overnight — whether through dual-listing of ATS operations, broader market-making obligations, or retail-directed liquidity programs — would reduce the overnight premium more effectively than steps aimed purely at information-quality improvements. For 24-hour-trading proponents, the results quantify the implicit tax on overnight transactions and indicate where investment in market infrastructure would have the largest welfare impact.

Several extensions suggest themselves. An examination of the 4pm-to-8pm post-close session, now coexisting with BOATS for some overlap hours, could separate the effects of the specific BOATS matching engine from the effects of overnight trading in general. Cross-asset extensions — comparing the adverse-selection signature of BOATS equity trading to the price dynamics of overnight options or single-stock futures — would inform debates on overnight derivatives market design. And a targeted study of overnight earnings-release events would quantify how much of the information premium accrues specifically to the handful of nights per quarter when each firm reports.

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